MILITARY SPECIFICATION

COMPASS SET, GYROSCOPE REFERENCE,
NAVIGATIONAL AN/ASN-75(*)

This amendment forms a part of Military Specification MIL-C-81412(WP) dated 17 March 1966 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment, basic MIL-C-81412(WP) is changed to MIL-C-81412(AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Page 2

3.2.2: Delete

3.3.1: Delete and substitute

"3.3.1 Total Weight - The total weight of the set, excluding cables, shall be a minimum consistent with good design and shall not exceed the following:

TYPE I

Item		Wt. (lbs.)
(1)	Gyroscope	4.2
(2)	Amplifier-Power Supply	1.8
(3)	Control	2.25
(4)	ML-1 Transmitter, Compass (with compensator)	1.3
	TOTAL	9.55

FSC 6605

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TYPE II

Item		Wt. (lbs.)
(1)	Gyroscope	4.2
(2)	Amplifier-Power Suqply	1.8
(3)	Control	2.25
(4)	ML-1 Transmitter, Compass (without compensator)	1.2
(5)	Compensator Electronic Magnetic Compass	1.1
	TOTAL	10.55"

Page 3

3.3.2.3: Delete and substitute

"3.3.2.3 Reliability in Specified Mean Time Between Failures (MTBF) - The equipment shall have a specified mean time between failures of 800 hours when tested and accepted as outlined under the requirements of 4.4.

Page 5

3.3.10, table: Add

Item	Gyroscope Displacement	Amplifier Power Supply	Control	Compensator Electronic
Vibration	Figure 2	Figure 3	Figure 4A	Figure 4

Page 6

3.4.1, third sentence: Delete and substitute

"The control azimuth signal shall be presented by a precision transmitting synchro as an output for indicators, navigational and accessory equipments, including compass adapters."

3.4.1, fourth sentence: Delete

3.4.2: Delete and substitute

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"3.4.2 Compass Set Accuracy -

TYPE I

Test	Standard Conditions	Envir. Extremes
Slaved Accuracy*	0.75° rms	1.0° rms
Gyro Drift	5.5°/hr.	9.5°/hr.
Slaving Rate	2.5° ±1.25°/min.	3.0° ±2.0°/min.

TYPE II

Test	Standard Conditions	Envir. Extremes
Slaved Accuracy*	0.25° rms	0.75° rms
Gyro Drift	5.5°/hr.	9.5°/hr.
Slaving Rate	2.5° ±1.25°/min.	3.0° ±2.0°/min.

* After index errors are removed. Includes ML-1 Transmitter Errors.

3.5.1, second and third sentences: Delete "transmitters" and substitute "transmitter".

Page 7

3.5.1.2: Delete "3.9" and substitute "4.2".

Page 8

3.5.3.3.2, first sentence: Delete and substitute

"The Control shall include a manually operated rotary knob and mechanism which shall have the capability of obtaining synchronization at a rate of at least 720 degrees per minute."

3.5.3.3.5: Delete "Type 2V-11 TRAC of MIL-S002-0708/103" and substitute

"Type 26V-11 TR4C of MIL-S-0020708/103"

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3.5.5.2: Delete "1.3" and substitute "1.1".

3.5.5.3: Delete "and contain mounting plate fasteners in accordance with MIL-F-25173".

Add the following new paragraphs:

* 3.5.6 Lubrication -

* 3.5.6.1 Rotor Bearing Lubrication - The rotor bearings shall be lubricated with grease conforming to MIL-G-81322 or MINIPURE grease with particle size not exceeding 50 microns.

* 3.5.6.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall be AS-2644 in accordance with the requirements of MIL-L-6085.

* 3.5.6.3 Barrier Film Bearing Coating - All bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.

* 3.5.6.4 Ball Bearings - Ball bearings shall be ABEC-5 grade or better. Teflon separators shall be used on gimbal bearings where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with lubricant AS-2844 and shall be ABEC-7 or better.

* 3.5.6.5 Ball Bearing Pre-Loading - Means shall be provided for preloading and alignment without causing eccentric loading unless otherwise authorized.

Page 12

4.4.1.3: Delete and substitute

"4.4.1.3 Synchronization - With the Remote Compass Transmitter set to north (0°) heading, the sync knob on the Control shall be rotated in the direction indicated on the panel until the annunciator on the Control is centered at 0°. The heading output from the Control shall indicate 0°. Rotation of the Compass Transmitter 10° clockwise from the synchronized position shall cause the annunciator to be fully deflected in the cross (+) direction (Fig. 7). Rotation of the "sync" knob in the direction indicated shall center the annunciator at 0°. The heading output signal from the Control shall indicate 10°. The Remote Compass Transmitter shall be repositioned to a north (0°) heading and the Compass Set synchronized. Rotation of the Compass Transmitter 10° counterclockwise from the synchronized

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position shall cause the annunciator to be fully deflected in the dot (·) direction (Fig. 7). Rotation of the "sync" knob in the direction indicated shall center the annunciator at 0°. The heading output signal from the Control shall indicate 350°."

4.4.1.4, first sentence: Delete "induction".

Page 13

4.4.1.5.2: Delete and substitute

"4.4.1.5.2 Scale Error - The scorsby shall be turned off during this test. The set shall be compensated using the Compensators provided in the set. Record the error between the Remote Compass Transmitter heading and the angle position indicator reading at 30° heading increments starting by rotating the sync knob on the controller. The set scale error shall be determined as follows:

$$\text{Scale Error} = \sqrt{(E^2)/R}$$

where E is the error at each heading after removal of index errors and R is the number of readings. The set scale error shall not exceed 0.75° standard conditions and 1.0° environmental conditions for the Type I set and 0.25° standard conditions and 0.75° environmental conditions for the Type II set."

4.4.1.6: Delete and substitute

"4.4.1.6 Gyro Drift - All gyro drift tests shall be conducted with the Slaved-Free switch in the "Free" position and with the gyro mounted on a 2° total excursion scorsby table oscillating at 5 to 7 revolutions per minute and reversing every 60 ±10 seconds. The scorsby table shall be initially leveled within 0.25° prior to any drift test."

Page 14

4.4.1.11, second sentence: Delete "(Figures 2 through 4)" and substitute

"and the applicable Figures 2 through 4A".

Page 15

4.4.1.18, line 11: Add

"except that the gyroscope amplitude shall be of ±2 g's."

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*4.4.3: Add the following to end of paragraph:

Classification of a failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1, last line: Delete "Test Plan II" and substitute "Test Plan IV"

Add the following new paragraph:

*4.4.3.1.1 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced but not submitted to the Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. Delivery of equipment shall not be held up to await a decision under the test plan. In event an Accept Decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan IV curves (extended) show a Reject Situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level E shall be used, except that the temperature extremes shall be +55C and -44C, and vibration shall be 0.5G. Details of the burn-in tests shall be included in the test procedures.

Page 22

6.10, under SPECIFICATIONS Military: Delete

"MIL-E-4682" and title

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Add the following under military specifications:

MIL-L-6085	Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range
MIL-B-81744	Barrier Coating Solution and Solvent

Add the following under military standards

MIL-STD-1334	Process for Barrier Coating of Anti-Friction Bearings

At end of para. 6.10, add the following:

Naval Air Systems Command

AR-34	Failure Classification for Reliability Testing, General Requirements for
AS-2844	Lubricant, Instrument Ball Bearing, High Flash Point

Page 24

Figure 1: Delete and substitute new Figure 1 of this amendment.

Page 27

Figure 4: Delete and substitute new Figure 4 of this amendment.

Page 27A

Add new Figure 4A of this amendment.

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Figure 11: Delete and substitute new Figure 11 of this amendment.

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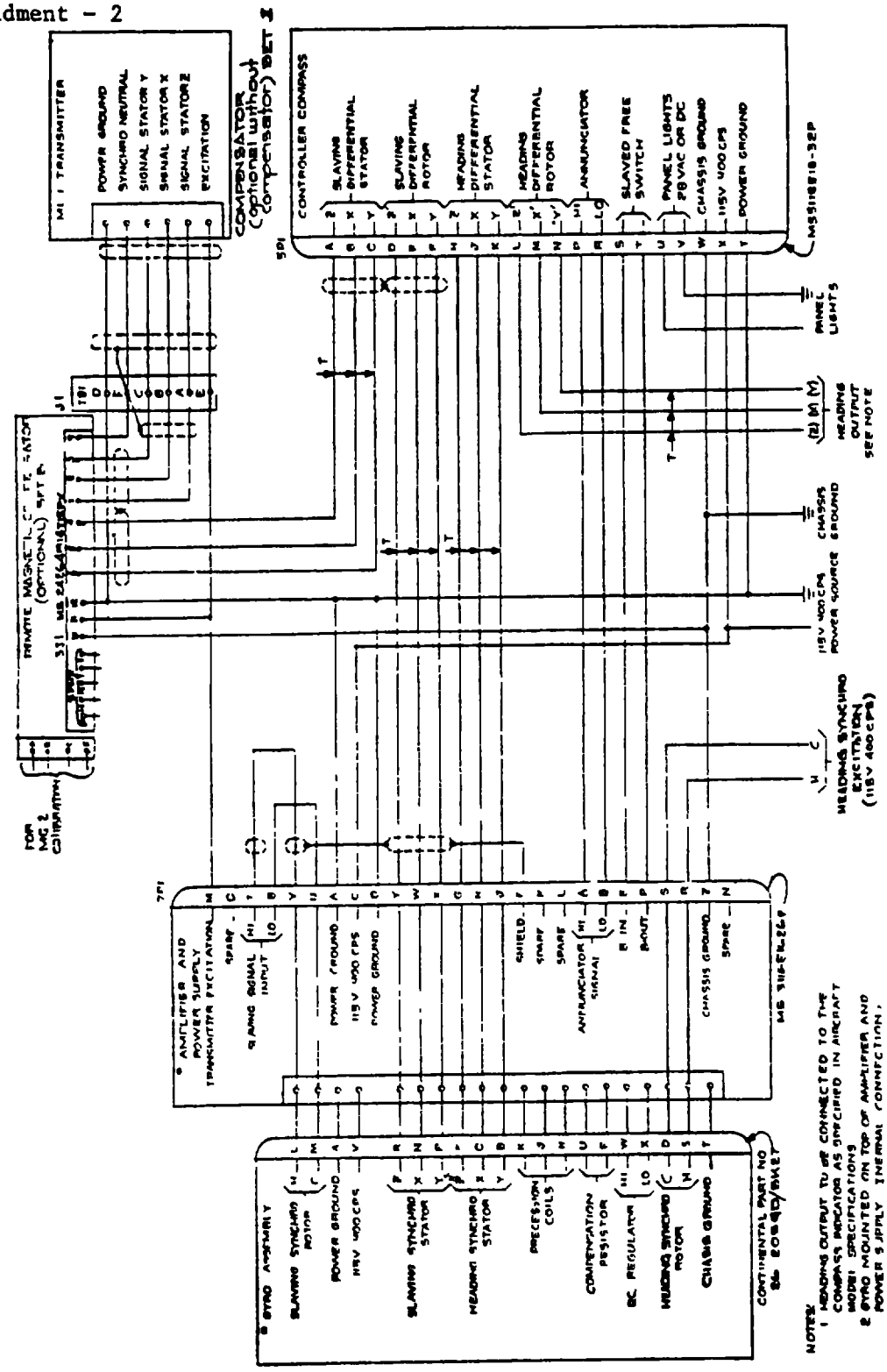


FIGURE 1

FIGURE 1
INTERCONNECTING WIRING FOR COMPASS SET, GYROSCOPE REFERENCE, NAVIGATIONAL AY/ASN 75

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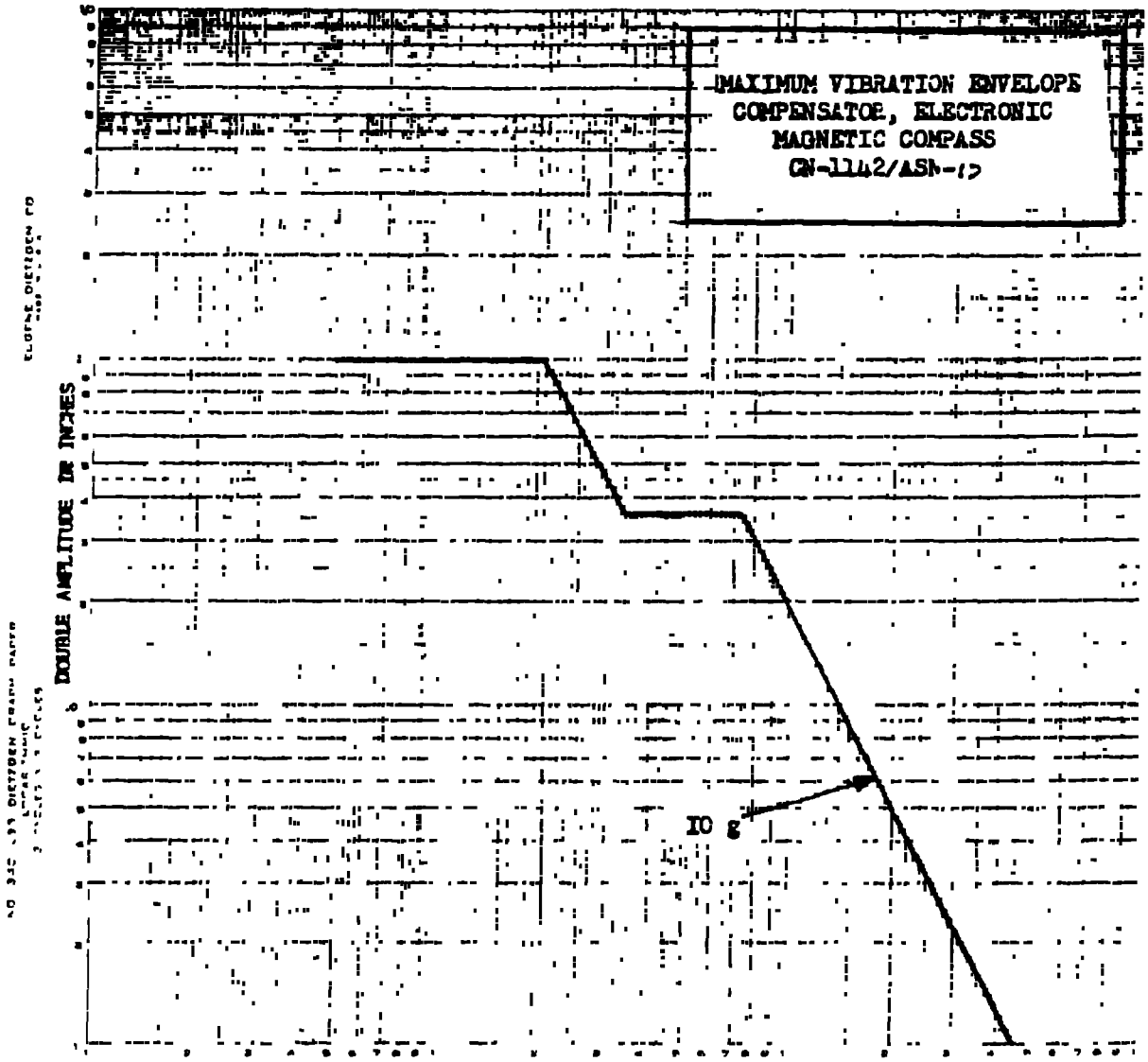


FIGURE 4.
CONTROL, COMPASS SET AND COMPENSATOR,
ELECTRONIC MAGNETIC COMPASS VIBRATION

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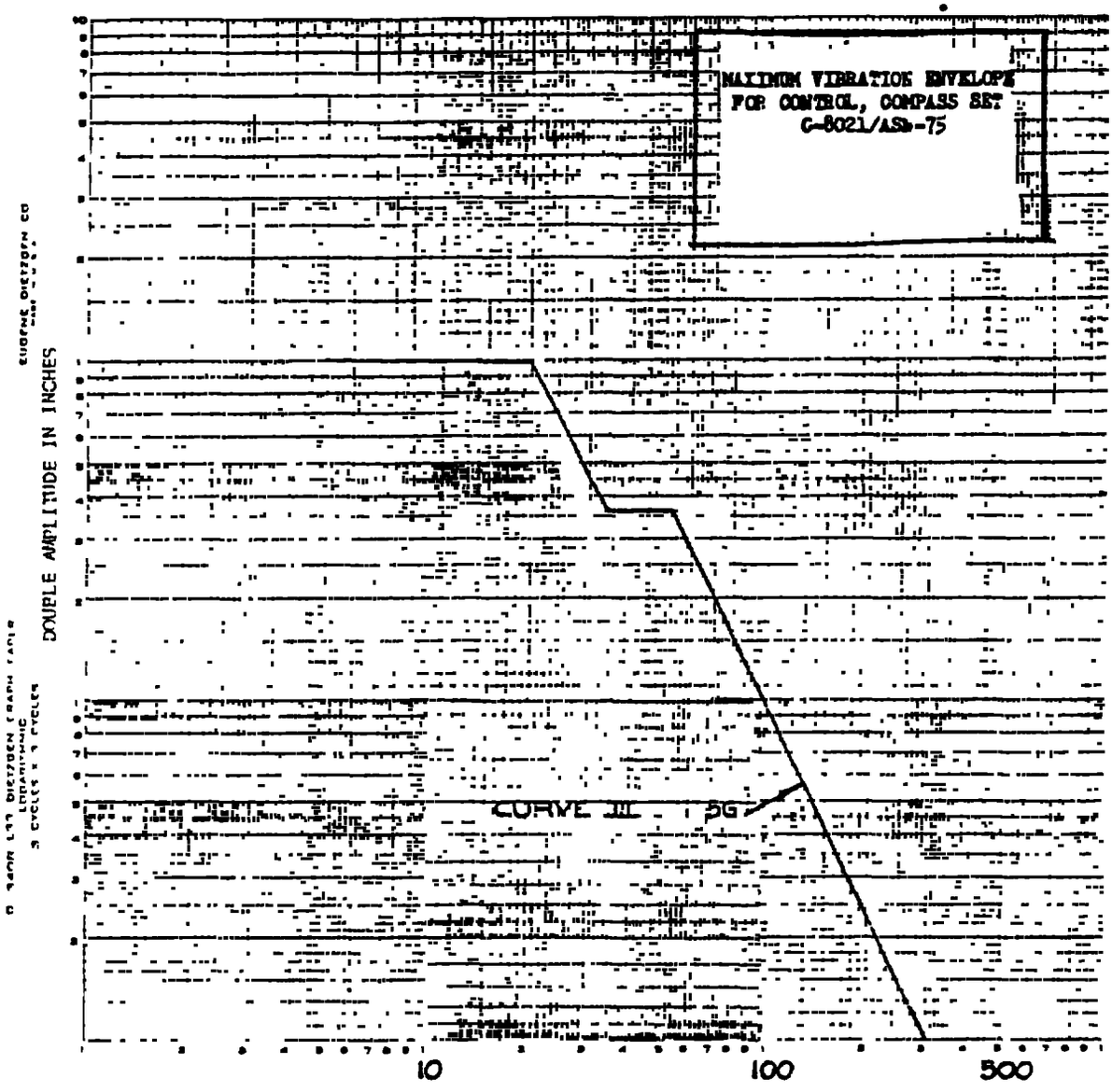


FIGURE 4A
CONTROL, COMPASS SET

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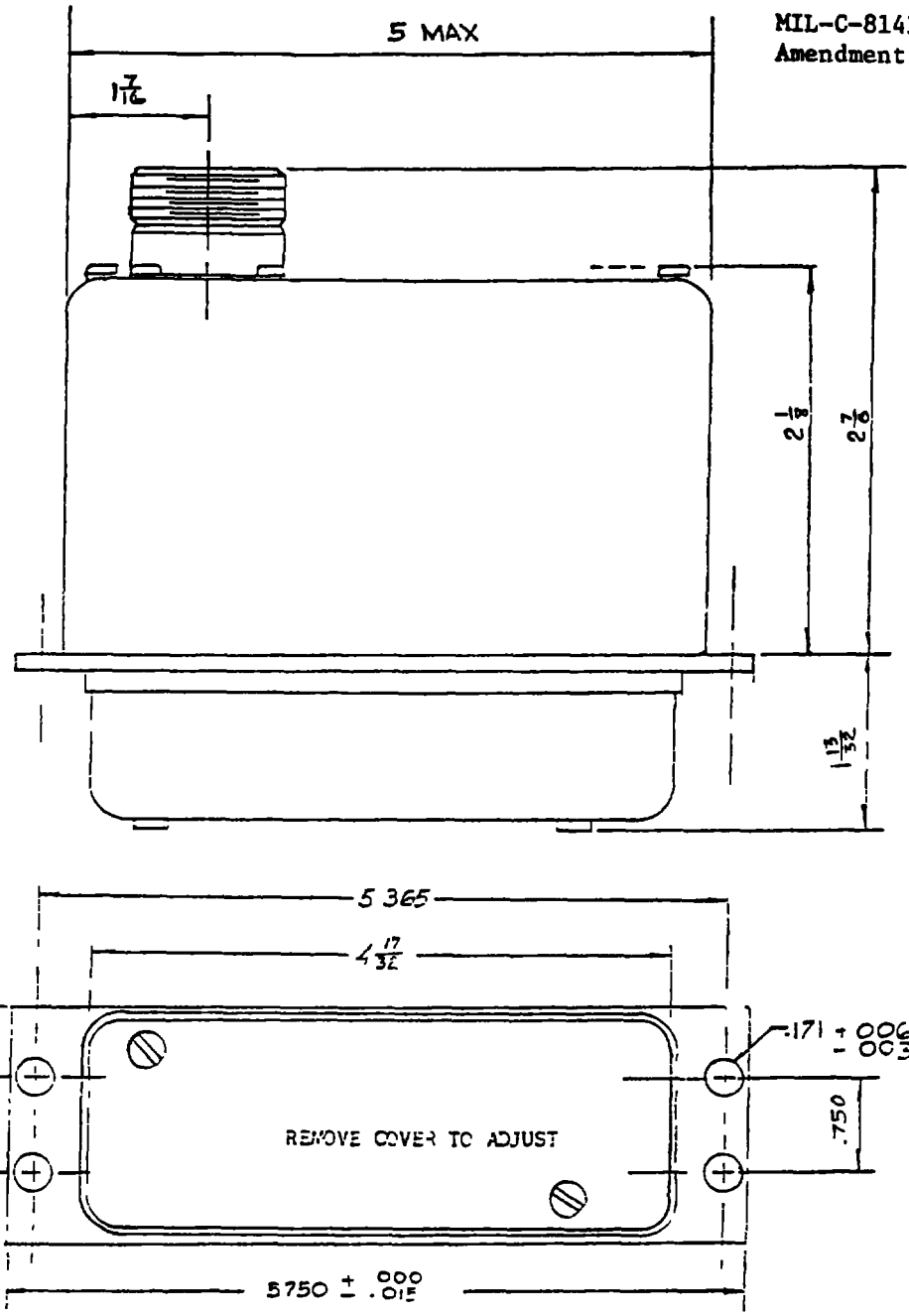


FIGURE 11

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